

ADAK, AK, USA

WMO: 704540

Lat: 51.883N

Lon: 176.650W

Elev: 17

StdP: 14.69

Time Zone: -10.00 (NAH)

Period: 94-19

WBAN: 25704

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.9	23.1	10.3	9.3	22.8	13.8	11.1	25.8	45.0	34.6	40.9	35.8	4.1	230	0.895

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.9	60.6	56.6	57.1	53.2	55.2	51.5	57.1	59.7	54.5	56.5	52.7	54.8	15.2	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
55.3	65.2	58.1	53.7	61.4	56.8	51.6	56.7	54.5	24.4	59.4	23.0	56.4	21.6	54.7	66.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
40.7	35.0	31.4	DB	12.6	64.9	3.7	2.6	9.9	66.8	7.8	68.3	5.7	69.7	3.1	71.6	
			WB	11.4	60.6	3.7	3.3	8.7	63.0	6.6	64.9	4.5	66.8	1.9	69.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	40.9	32.6	34.0	34.3	36.9	40.8	45.8	49.6	51.9	48.9	43.3	38.1	34.0
	DBStd	7.60	5.42	4.34	4.23	2.96	2.67	2.46	2.71	3.11	3.29	3.74	4.07	4.68
	HDD50	3454	540	449	487	392	286	129	39	11	56	211	357	497
	HDD65	8803	1005	869	952	842	751	577	477	405	482	674	807	962
	CDD50	126	0	0	0	0	0	2	27	71	24	2	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
	WSAvg	14.3	14.5	17.1	16.0	14.9	13.3	11.6	11.0	11.6	14.6	15.2	15.8	16.2
	PrecAvg	56.5	5.4	4.0	5.0	4.2	3.1	2.5	2.8	4.0	5.1	6.0	7.2	6.7
Precipitation	PrecMax	69.8	10.3	8.2	9.6	10.2	6.0	4.9	6.1	7.3	11.3	10.0	11.7	10.2
	PrecMin	37.4	2.7	2.0	2.3	1.4	0.6	0.9	0.7	1.3	2.5	2.6	3.0	3.9
	PrecStd	8.2	1.7	1.5	1.8	1.9	1.3	0.9	1.3	1.5	1.9	1.9	2.3	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.4	45.3	45.2	46.1	51.0	56.5	63.0	65.0	60.7	54.6	51.8	46.7
		MCWB	41.6	42.4	42.3	42.2	45.3	51.1	56.9	62.0	55.7	49.8	49.0	43.8
	2%	DB	43.1	43.1	43.2	44.6	47.4	53.7	57.5	61.1	56.8	52.2	47.9	44.7
		MCWB	39.7	40.0	40.1	40.7	43.4	48.6	53.2	57.6	52.5	48.0	45.9	41.9
	5%	DB	41.1	41.1	41.2	43.1	45.8	51.8	55.2	58.6	55.0	50.2	45.9	41.4
		MCWB	38.4	38.3	38.2	39.2	41.6	47.4	51.4	55.1	51.2	46.7	43.6	38.6
	10%	DB	39.2	39.3	39.4	41.4	45.1	49.4	54.2	56.5	54.2	48.3	44.9	39.4
		MCWB	36.8	36.4	36.4	38.1	41.0	46.0	50.5	52.9	50.4	45.1	42.0	36.7
	0.4%	WB	42.8	43.5	43.6	44.3	46.5	51.7	58.6	62.4	57.6	51.0	49.2	45.6
		MCDB	44.8	45.4	45.0	45.3	47.7	55.0	61.6	64.4	59.2	53.4	50.7	46.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	40.7	40.9	41.1	41.3	44.3	49.1	54.6	58.6	54.2	48.5	46.2	41.8
		MCDB	42.9	42.7	43.1	43.6	46.2	52.2	57.4	61.0	55.7	51.1	47.2	44.3
	5%	WB	39.0	39.0	39.1	40.2	42.6	47.3	52.3	56.0	52.4	46.7	44.4	39.4
		MCDB	41.0	40.9	41.2	42.6	45.1	50.1	55.1	58.3	54.3	49.3	45.7	41.6
	10%	WB	37.5	37.4	37.6	38.8	41.5	46.2	50.5	53.6	50.5	45.9	42.0	37.9
		MCDB	39.1	39.2	39.6	41.3	44.3	48.4	53.3	55.4	53.1	47.9	44.6	39.8
Mean Daily Temperature Range	5% DB	MDBR	7.7	6.8	7.1	7.5	7.0	6.9	7.5	7.9	8.0	7.9	7.1	7.6
		MCDBR	7.1	7.6	8.4	8.8	8.5	10.4	10.6	10.9	9.4	9.1	9.2	8.2
		MCWBR	6.5	7.7	7.5	7.3	6.4	7.1	7.8	8.4	6.9	7.5	9.2	8.0
	5% WB	MDBR	6.8	7.9	8.3	8.1	7.6	8.8	10.5	10.3	8.3	8.5	9.2	8.2
		MCDBR	6.6	8.2	8.2	7.3	6.2	6.7	8.4	8.6	7.1	7.6	9.7	8.2
		MCWBR												
Clear-Sky Solar Irradiance	taub		0.312	0.334	0.358	0.435	0.428	0.416	0.431	0.402	0.391	0.354	0.344	0.304
	taud		2.318	2.311	2.293	2.041	2.133	2.260	2.264	2.366	2.291	2.310	2.195	2.289
	Ebn at Noon		217	246	263	252	262	266	259	262	248	234	194	197
	Edh at Noon		20	27	33	48	46	41	40	35	33	27	23	18
All-Sky Solar Radiation	RadAvg		225	413	786	1164	1237	1241	1190	1034	861	591	279	178
	RadStd		20	47	65	102	98	101	97	96	76	55	26	15

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page